

Credit Risk Decisioning Technical Exercise

Thank you for considering this technical exercise as part of our selection process. The objective of this task is to assess your ability to work across Credit Risk Analytics, Credit Risk Modelling, and Model Deployment in a small team setting. The exercise is designed to be deliberately open-ended, giving you the opportunity to demonstrate both your technical skills and thought process.

Task Overview

You are provided with a snapshot of a consumer loan portfolio as of April 2019, including all originations from 2007 up to that date. Your goal is to explore the dataset, derive insights, build a credit risk model, and package the model for deployment. The expected deliverables are three **Jupyter Notebooks**, each focusing on a specific stage:

1. **Analytics**
 - Define what constitutes a “bad” outcome based on the dataset.
 - Explore how key variables impact the likelihood of bad outcomes.
 - Assess how historical credit risk policies have shaped the current portfolio composition.
 - Provide any additional insights that could influence credit risk decisioning.
2. **Modelling**
 - Build and validate a credit risk model using the dataset of accepted applications.
 - Discuss the model’s performance and whether it has uncovered additional trends or hidden features beyond the initial analytics.
 - Apply the model to rejected applications (if possible) and discuss how they would score.
 - Suggest ways the credit risk policy could be refined based on your model’s outcomes.
3. **Model Deployment**
 - Package your model into a reusable function that can be imported from a local helper file.
 - Implement an inference function that takes application data as input and returns a score in a few lines of code.
 - Structure the inference function in a way that could be easily adapted to an **AWS Endpoint** deployment (similar to an `inference.py` format).

Timeline

We expect the completed solution to be submitted **within one week** from today. We recognise this is a tight timeframe, so we encourage prioritising a **working, end-to-end solution** covering all three stages rather than spending excessive time on fine-tuning any single aspect. You will have the opportunity to discuss possible improvements during the debrief session.

Use of AI Tools

You are free – and encouraged – to use **LLM-generated assistance** as long as:

- The suggestions do not rely on obscure or difficult-to-install Python packages.
- You can confidently defend any AI-generated code during the debrief.

Dataset

- Dataset: [Lending Club Loan Data on Kaggle](#)
- Data Dictionary: [LC Data Dictionary](#)

It is up to you to determine which data fields are suitable and how best to leverage them.

We look forward to seeing how you approach this challenge!